$$I = \frac{e^{ax}}{a}\sin bx - \frac{b}{a}\frac{e^{ax}}{a}\cos bx - \frac{b^2}{a^2}\int e^{ax}\sin bx\,dx$$

$$I = \frac{e^{ax}}{a}\sin bx - \frac{b}{a^2}e^{ax}\cos bx - \frac{b^2}{a^2}I$$

$$\left(1+\frac{b^2}{a^2}\right)I = \frac{ae^{ax}\sin bx - be^{ax}\cos bx}{a^2}$$

$$\left(\frac{a^2+b^2}{a^2}\right)I = \frac{e^{ax}[a\sin bx - b\cos bx]}{a^2}$$

$$\text{Therefore, } I = \frac{e^{ax}}{a^2+b^2}[a\sin bx - b\cos bx]+c$$

$$\text{Therefore, } \int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\sin bx - b\cos bx]+c$$

$$\text{Similarly, } \int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\cos bx + b\sin bx]+c$$

$$\boxed{\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\sin bx - b\cos bx]+c}$$

$$\boxed{\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\cos bx + b\sin bx]+c}$$

Caution

In applying integration by parts to specific integrals, the pair of choice for u and dv once initially assumed should be maintained for the successive integrals on the right hand side. (See the above two examples). The pair of choice should not be interchanged.

Examples 11.36

Evaluate the following integrals

(i) $\int e^{3x}\cos 2x\,dx$ (ii) $\int e^{-5x}\sin 3x\,dx$

(i) $\int e^{3x}\cos 2x\,dx$

Using the formula

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\cos bx + b\sin bx]+c$$

For $a=3$ and $b=2$, we get

$$\int e^{3x}\cos 2x\,dx = \left(\frac{e^{3x}}{3^2+2^2}\right)(3\cos 2x + 2\sin 2x)+c$$

$$= \left(\frac{e^{3x}}{13}\right)(3\cos 2x + 2\sin 2x)+c.$$

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